

Disagreement-Aware Multi-Agent Investment Research Decision System under Financial Uncertainty

ABSTRACT

Master thesis: XX p, 6 figures, 4 tables, 24 sources.

LARGE LANGUAGE MODEL, MULTI-AGENT SYSTEM, FINANCIAL DECISION-MAKING, RETRIEVAL-AUGMENTED GENERATION, EVIDENCE CREDIBILITY, DISAGREEMENT DIAGNOSIS, RISK CONTROL, REVIEW MEMORY

This thesis studies a multi-agent investment research decision system for financial markets under uncertainty. The work is based on the Trading-Agents-OpenClaw project and improves it with three mechanisms: credibility-weighted retrieval-augmented generation, multi-dimensional disagreement diagnosis, and review-memory-driven self-calibration.

Financial decision-making is difficult because price data, fundamentals, news, sentiment, and macro events may point in different directions. A single large language model can summarize such information, but it may also overreact to noisy evidence, ignore conflicting signals, or repeat past mistakes. The proposed system treats disagreement among financial agents not as a by-product of discussion, but as an observable signal of uncertainty. It converts market, fundamental, sentiment, news, risk, and trading outputs into structured records, estimates evidence credibility, measures disagreement, and then adjusts the final trading action and position size.

Experiments are conducted using the project scripts and cached multi-agent outputs. In the mechanism-level ablation study, the full model obtains a disagreement score of 0.466 and limits the suggested position to 20.57%, while the model without disagreement diagnosis sets disagreement to zero and increases the position to 30.00%. In the four-stock backtest with 20 samples, the full model has a total return of -0.59% and a maximum drawdown of 2.01%, while the three ablated models produce larger losses and drawdowns. The results show that the proposed uncertainty-aware mechanisms improve interpretability and reduce risk exposure in high-conflict cases.

CONTENTS

1 Introduction

1.1 Research Background

Financial markets are dynamic systems in which structured data and unstructured information interact. Prices are affected by corporate fundamentals, technical patterns, liquidity, macro policy, investor sentiment, and unexpected events. Classical financial analysis and quantitative trading methods can be efficient when their assumptions are stable, but they are often weak at interpreting heterogeneous text evidence and explaining conflicting signals [1–4].

Large language models have expanded the ability of software systems to read financial reports, summarize news, call tools, and produce natural-language reasoning. Financial language models such as FinBERT, BloombergGPT, and FinGPT demonstrate that domain-specific language understanding is useful for financial text and decision support [7–9]. At the same time, agent frameworks such as ReAct, AutoGen, CAMEL, and Generative Agents show that language models can be organized into tool-using and memory-augmented agents [11–14].

TradingAgents applies this idea to investment research by simulating a trading firm with market analysts, fundamental analysts, news analysts, bull and bear researchers, risk managers, and traders [18]. The Trading-Agents-OpenClaw project extends this workflow with local scripts, data integration, reporting, and review functions. However, the original workflow still leaves three issues open. First, agent disagreement is mostly expressed as text and is not directly used for risk control. Second, retrieved financial evidence is not sufficiently weighted by reliability. Third, historical review is often stored as a narrative rather than as a calibration signal for later decisions.

1.2 Aim and Contributions

The aim of this thesis is to design and implement an uncertainty-aware multi-agent investment research system. The system must preserve the explainability of

the original multi-agent workflow while making disagreement and evidence quality computable.

The main contributions are:

- 1) A credibility-weighted RAG mechanism that scores financial evidence by source authority, freshness, cross-source consistency, data quality, and historical reliability.
- 2) A disagreement diagnosis mechanism that measures direction, confidence, evidence, risk, and time-horizon disagreement among financial agents.
- 3) A risk-adaptive decision mechanism that maps disagreement into position reduction and control actions.
- 4) A review memory mechanism that records decisions and market feedback for later calibration.
- 5) An implementation and ablation evaluation based on the Trading-Agents-OpenClaw project.

1.3 Research Scope

The system studied in this thesis is not intended to be a fully automated trading robot. It is positioned as a decision-support system for investment research. This distinction is important. A trading robot would need order execution, broker integration, strict compliance controls, capital allocation rules, and real-time fail-safe mechanisms. The present work focuses on the research and decision layer: collecting evidence, organizing agent opinions, detecting uncertainty, and producing a risk-adjusted recommendation.

The research scope is therefore limited to three questions. First, how can heterogeneous financial information be represented in a form that is suitable for both natural-language reasoning and quantitative control? Second, how can disagreement among multiple financial agents be measured and interpreted? Third, how can this disagreement influence final position size rather than remaining only a textual explanation in a report?

The system assumes that external data interfaces can provide the required market and textual information. It also assumes that the large language model is used as a reasoning component rather than as a guaranteed forecasting oracle. The output of the system should be interpreted as an auxiliary research signal. Human review and independent risk control remain necessary.

1.4 Thesis Organization

The thesis is organized as follows. Section 2 reviews financial language models, LLM agents, retrieval-augmented generation, and financial risk evaluation. Section 3 describes the system design and explains how the proposed modules are mapped to project files. Section 4 presents the core method, including evidence credibility scoring, disagreement diagnosis, risk-adaptive position control, and review memory. Section 5 reports implementation details and experimental results. Section 6 summarizes the work and discusses future directions.

2 Related Work

2.1 Financial Language Models

FinBERT adapts BERT to financial sentiment analysis and shows the value of domain-specific language modeling [6, 7]. BloombergGPT trains a large language model on broad financial corpora, while FinGPT emphasizes an open-source and data-centric approach for financial large language models [8, 9]. These works focus mainly on language understanding and generation. This thesis focuses on how financial language outputs can be organized into a risk-aware multi-agent decision process.

Financial language differs from general news text in several ways. First, it uses compact domain terminology, such as earnings surprise, free cash flow, valuation multiple, drawdown, and guidance revision. Second, the meaning of a sentence is often conditional on time. A report that is useful before an earnings release may become outdated immediately after the release. Third, financial text often contains mixed evidence. A company may report strong revenue growth but weak margins, or a sector may benefit from policy support while facing short-term liquidity pressure. These characteristics make financial language modeling useful but also risky when the system lacks explicit evidence control.

For this reason, the thesis does not attempt to train a new financial language model. Training a domain model would require a large corpus, substantial computing resources, and a separate evaluation protocol. Instead, this work assumes that a capable LLM already exists and studies how its outputs can be made more reliable through system design. The proposed mechanisms are model-agnostic: the same evi-

dence scoring and disagreement diagnosis can be used with different LLM providers, provided that the agent outputs can be converted into structured fields.

2.2 LLM Agents and Multi-Agent Collaboration

ReAct combines reasoning traces and tool actions, which is important for financial systems that must retrieve data and justify decisions [11]. AutoGen and CAMEL provide general multi-agent conversation frameworks [12, 13]. Generative Agents, Reflexion, Self-Refine, and Voyager demonstrate that memory, feedback, and iterative improvement can improve agent behavior [14–17]. This thesis adopts these ideas in a financial setting, where feedback is delayed and risk-sensitive.

The multi-agent design is especially suitable for financial research because real investment decisions are rarely made from a single viewpoint. A technical analyst may focus on trend and volume, a fundamental analyst may focus on profitability and valuation, a news analyst may focus on events, and a risk manager may focus on downside scenarios. In a language-agent system, these roles can be represented by separate prompts and separate tool contexts. The benefit is not only parallel analysis, but also disagreement exposure.

However, multi-agent systems also introduce a new problem. If each agent produces a long natural-language report, the final trader agent may be forced to summarize conflicting opinions without knowing how severe the conflict is. A bullish fundamental report and a bearish technical report may both be plausible. If the system simply asks another LLM to decide, it may hide the conflict behind a confident final answer. The main idea of this thesis is to keep the conflict visible and computable.

Therefore, the proposed workflow separates two tasks. Natural-language agents still produce rich reasoning and evidence descriptions. A structured uncertainty layer then extracts action, confidence, evidence stance, risk points, and time horizon from these outputs. This layer does not replace the agents; it converts their disagreement into a risk-control signal.

2.3 Retrieval-Augmented Generation

Retrieval-augmented generation combines a parametric language model with an external document store [10]. It is useful in finance because current news and filings change faster than model parameters. However, standard RAG usually ranks

documents by relevance, while financial decisions also require credibility. Therefore, this thesis adds explicit evidence scoring and conflict detection.

In a general question-answering task, the most relevant document is often the best context. In finance, this assumption is too weak. A highly relevant social-media rumor may be less useful than a less semantically similar but official exchange disclosure. Similarly, a recent news article may be more important for a short-term trade than a three-month-old commentary. The proposed credibility-weighted RAG mechanism therefore treats retrieval as a two-dimensional problem: relevance determines whether a document is connected to the query, while credibility determines how strongly it should influence the reasoning process.

The mechanism also handles evidence conflict. If one evidence item supports a bullish conclusion while another supports a bearish conclusion, the system records this conflict rather than forcing an immediate merge. The conflict is then passed to the disagreement layer, where it contributes to the evidence disagreement component.

2.4 Financial Risk Evaluation

Financial strategies are commonly evaluated by return, volatility, maximum drawdown, win rate, and risk-adjusted return [3, 5]. Deep reinforcement learning frameworks such as FinRL highlight the importance of reproducible backtesting and practical constraints in trading research [19–21]. This thesis uses a lightweight backtest to verify whether the proposed mechanisms reduce risk exposure.

The evaluation in this thesis is intentionally conservative. The backtest is not used to claim that the system can consistently outperform the market. The sample is too small for that claim. Instead, the experiment asks a narrower question: when the same raw multi-agent analysis is processed by different uncertainty modules, does the full system take more cautious positions in high-disagreement cases, and does this reduce losses in the tested samples?

This evaluation logic matches the goal of the thesis. The main target is not price prediction accuracy alone, but risk-aware decision control. A model that produces a correct direction but excessive position size may still be dangerous. Conversely, a system that lowers exposure under high uncertainty may be useful even when the final return remains slightly negative in a short sample.

2.5 Summary of Related Work

The reviewed literature supports three design principles. First, financial language processing benefits from domain awareness, but language understanding alone is not sufficient for risk control. Second, LLM agents are effective at decomposing complex tasks, but multi-agent disagreement must be measured explicitly. Third, financial evaluation should include both return and risk indicators. The proposed system combines these principles in a project-level implementation.

3 System Design

3.1 Project Basis

The implemented system is based on Trading-Agents-OpenClaw. The relevant project files are summarized in Table 1.

Table 1 – Mapping between thesis modules and project files

Thesis module	Project file or directory	Function
Analysis entry	<code>run_analysis.py</code>	Runs multi-agent analysis and integrates uncertainty modules
Disagreement diagnosis	<code>disagreement.py</code>	Computes multi-dimensional disagreement
Adaptive decision	<code>adaptive_decision.py</code>	Converts disagreement into risk-aware action and position
Evidence weighting	<code>evidence.py</code>	Scores evidence credibility and identifies conflicts
Review memory	<code>review_memory.py</code> ; <code>review_update.py</code>	Stores feedback and updates calibration variables
Validation	<code>run_backtest.py</code> ; <code>run_ablation.py</code>	Runs ablation and backtest experiments

3.2 Functional Requirements

The system is designed around the investment research workflow. The first requirement is multi-source data acquisition. The system should collect historical prices, technical indicators, financial statements, company announcements, news, and sentiment signals. These inputs have different formats and reliability levels, so the system must not treat them as interchangeable text fragments.

The second requirement is evidence structuring. Each useful piece of information should be converted into an evidence record with fields such as source, time, type, stance, relevance, credibility, and conflict identifiers. This structure allows later modules to trace why a decision was made and which evidence items contributed to it.

The third requirement is multi-agent analysis. The system should allow specialized agents to analyze the same ticker from different viewpoints. Each agent should output not only a report, but also structured fields: decision direction, confidence, evidence identifiers, risk points, suggested position, and time horizon.

The fourth requirement is uncertainty-aware decision control. The final action must not be based only on a textual summary. It should also depend on measured disagreement. When disagreement is high, the system should reduce position size, request additional evidence, trigger a second debate, or keep the recommendation conservative.

The fifth requirement is reviewability. Every analysis should be saved in a form that can be checked later. This includes raw agent reports, structured outputs, evidence scores, disagreement scores, final action, position, and later market feedback.

3.3 Non-Functional Requirements

The first non-functional requirement is explainability. A financial decision system must explain why it recommends buy, hold, or sell. The explanation should include key evidence, agent opinions, disagreement sources, and risk warnings.

The second requirement is robustness. Financial information is noisy, and the system should avoid overreacting to isolated signals. Robustness is improved by credibility scoring, cross-source checking, and position reduction under high uncertainty.

The third requirement is modularity. Evidence scoring, disagreement diagnosis, adaptive decision, and review memory should be implemented as separate modules. This makes it possible to test each mechanism independently and replace one component without rewriting the whole system.

The fourth requirement is reproducibility. For thesis experiments, repeated LLM calls can introduce noise. Therefore, the ablation experiment uses cached raw outputs. This ensures that differences between variants are caused by the uncertainty modules rather than random generation changes.

3.4 System Architecture

The system contains five layers.

- 1) The data layer collects market prices, technical indicators, financial data, news, and sentiment.
- 2) The evidence layer converts raw material into structured evidence records.
- 3) The agent layer runs market, fundamental, sentiment, news, risk, and trading agents.
- 4) The uncertainty layer computes disagreement and controls the final action.
- 5) The review layer records decisions and later feedback.

The key design choice is to keep the natural-language reasoning of the agents but extract structured fields from it. Each agent output includes action, confidence, evidence identifiers, suggested position, risk points, and time horizon.

3.5 Core Data Structures

The evidence object is the basic unit of the evidence layer. It contains the evidence identifier, ticker, source, publication time, evidence type, content summary, stance, relevance score, credibility score, and conflict identifiers. This object makes evidence traceable and prevents the system from losing the source of a claim after it is inserted into an LLM prompt.

The agent output object contains the agent name, role, action, confidence, evidence identifiers, risk points, suggested position, reasoning summary, and time horizon. This object connects natural-language reports with quantitative disagreement calculation.

The decision result object stores the final action, final position, disagreement

score, main disagreement type, risk level, key evidence, and explanation. It is used both for report generation and for review memory.

The review memory object stores realized return, drawdown, decision correctness, effective agents, misleading agents, effective evidence, misleading evidence, and lessons. It is not a replacement for model training. It is an external memory that adjusts weights and thresholds in later runs.

3.6 Workflow

The workflow begins with a ticker and an analysis date. The system collects raw data and converts it into candidate evidence. The evidence layer scores each item and detects conflicts. The multi-agent layer then generates structured reports. The uncertainty layer computes disagreement across agent outputs and evidence stances. Finally, the adaptive decision module produces the action and position, and the review memory module stores the decision for later feedback.

This workflow keeps the advantages of the original TradingAgents design while adding an explicit control loop. The original agents remain responsible for domain reasoning. The new modules are responsible for reliability and risk control.

4 Core Method

4.1 Evidence Credibility

Let e_j be a candidate evidence item. Its credibility score is defined as

$$C(e_j) = \alpha_s S_j + \alpha_t T_j + \alpha_c C_j + \alpha_q Q_j + \alpha_h H_j, \quad (1)$$

where S_j is source authority, T_j is freshness, C_j is cross-source consistency, Q_j is data quality, and H_j is historical reliability. The weights satisfy

$$\alpha_s + \alpha_t + \alpha_c + \alpha_q + \alpha_h = 1, \quad \alpha_s, \alpha_t, \alpha_c, \alpha_q, \alpha_h \geq 0. \quad (2)$$

In the initial setting, the weights are

$$\alpha_s = 0.25, \quad \alpha_t = 0.20, \quad \alpha_c = 0.20, \quad \alpha_q = 0.20, \quad \alpha_h = 0.15. \quad (3)$$

The final retrieval priority combines relevance and credibility:

$$W(e_j) = \beta R(e_j) + (1 - \beta)C(e_j), \quad 0 \leq \beta \leq 1. \quad (4)$$

The scoring function is intentionally simple. It is not meant to represent a complete theory of financial truth. Instead, it gives the system a transparent and adjustable way to prefer stronger evidence. For example, an official filing may receive a high source authority score, while a social-media post may receive a low score. A recent article may receive a high freshness score for a short-term analysis, while old news may be downgraded. If multiple independent sources support the same stance, the cross-source consistency score increases.

The credibility score also affects prompt construction. High-credibility evidence is placed earlier in the context and may receive a longer summary. Low-credibility evidence is either shortened or marked as auxiliary information. Conflicting evidence is not removed automatically, because removing it may hide important uncertainty. Instead, conflict information is preserved and passed to the disagreement module.

4.2 Evidence Conflict Detection

Evidence conflict is detected by comparing the stances of evidence items. Let $s_j \in \{-1, 0, 1\}$ denote the stance of evidence item e_j , where -1 means bearish, 0 means neutral, and 1 means bullish. Two evidence items are considered directionally conflicting when

$$s_j s_k < 0. \quad (5)$$

The conflict indicator is then stored in the evidence object:

$$\text{conflict}(e_j, e_k) = \begin{cases} 1, & s_j s_k < 0, \\ 0, & \text{otherwise.} \end{cases} \quad (6)$$

This conflict rule is deliberately interpretable. It can be extended in future work by using semantic contradiction detection, entailment models, or event-level relation extraction.

4.3 Disagreement Diagnosis

For n agents, each agent i outputs a decision $a_i \in \{-1, 0, 1\}$ for sell, hold, and buy. The weighted direction score is

$$\bar{a} = \sum_{i=1}^n w_i a_i, \quad \sum_{i=1}^n w_i = 1. \quad (7)$$

Direction disagreement is measured as weighted variance:

$$D_{\text{dir}} = \sum_{i=1}^n w_i (a_i - \bar{a})^2. \quad (8)$$

Confidence disagreement is defined by the dispersion of confidence scores:

$$D_{\text{conf}} = \sqrt{\frac{1}{n} \sum_{i=1}^n (q_i - \bar{q})^2}. \quad (9)$$

The total disagreement score is

$$D = \lambda_1 D_{\text{dir}} + \lambda_2 D_{\text{conf}} + \lambda_3 D_{\text{evd}} + \lambda_4 D_{\text{risk}} + \lambda_5 D_{\text{hor}}, \quad (10)$$

where

$$\lambda_1 + \lambda_2 + \lambda_3 + \lambda_4 + \lambda_5 = 1, \quad \lambda_k \geq 0. \quad (11)$$

The five disagreement components have different meanings. Direction disagreement measures whether agents recommend different actions. Confidence disagreement measures whether agents express different levels of certainty. Evidence disagreement measures whether the evidence used by agents points in different directions. Risk disagreement measures whether risk warnings are consistent. Horizon disagreement measures whether agents disagree because they analyze different time scales.

This decomposition is useful because different conflicts require different control actions. A direction conflict may require a second debate. An evidence conflict may require re-retrieval. A risk conflict may require risk-manager arbitration. A horizon conflict may not mean that one agent is wrong; it may indicate that short-term and medium-term strategies should be separated.

4.4 Control Action Mapping

The system maps the main disagreement type to a control action. If D is low, the final decision is allowed to follow the trading agent with normal position control. If D is medium, the system reduces position size. If D is high, the system may recommend hold or request additional evidence before action. This mapping is summarized as follows:

$$u(D) = \begin{cases} \text{normal}, & D < \tau_1, \\ \text{reduce}, & \tau_1 \leq D < \tau_2, \\ \text{review}, & D \geq \tau_2. \end{cases} \quad (12)$$

The thresholds τ_1 and τ_2 can be calibrated by review memory. In the current implementation, the threshold logic is rule-based, which makes it easy to inspect and debug.

4.5 Risk-Adaptive Position

The final position is reduced when disagreement is high:

$$p = \min(p_{\max}, \max(0, p_0(1 - \gamma D)m_a)), \quad (13)$$

where p_0 is the base position, γ is the disagreement penalty, m_a is an action multiplier, and $p_{\max}$ is the maximum allowed position.

Equation 13 is the core connection between disagreement and trading exposure. If the agents agree and D is small, the position remains close to the base value. If disagreement increases, the term $(1 - \gamma D)$ reduces exposure. The maximum and minimum operators prevent invalid positions.

This design follows a conservative assumption: when the system cannot reach a stable consensus, it should reduce exposure rather than increase confidence. This does not mean that disagreement is always negative. Sometimes disagreement may reveal an opportunity. However, for a thesis-level decision-support system, it is safer to first treat high disagreement as a risk signal.

4.6 Review Memory

After a decision, the system records the ticker, date, agent outputs, evidence scores, disagreement score, final action, position, realized return, and drawdown. A simplified memory update is

$$\theta_{t+1} = (1 - \eta)\theta_t + \eta g(r_t, d_t), \quad (14)$$

where θ_t denotes calibration parameters, r_t is realized return, d_t is drawdown, and η is the update rate. This formula does not update language-model parameters. It updates external weights and thresholds used by the decision pipeline.

Review memory has three roles. First, it stores an audit trail. A user can inspect what evidence and agent opinions led to a recommendation. Second, it supports later calibration. If an evidence source repeatedly leads to misleading conclusions, its historical reliability can be reduced. Third, it helps identify systematic errors, such as excessive position size during high disagreement or overreliance on one type of evidence.

The current implementation uses JSON-based memory records. This is simple and transparent, but not ideal for large-scale deployment. In future work, the memory could be stored in a database with query support, time decay, and market-regime tags.

5 Implementation and Experiments

5.1 Experimental Setup

The experiments compare four configurations:

- 1) Ours-full: credibility-weighted RAG, disagreement diagnosis, adaptive decision, and review memory enabled.
- 2) w/o CW-RAG: evidence credibility weighting disabled.
- 3) w/o DADM: disagreement-aware decision module disabled.
- 4) w/o RMSE: review-memory self-evolution disabled.

To reduce randomness from repeated LLM calls, the ablation study uses cached raw multi-agent outputs. Each variant is evaluated on the same underlying analysis result.

The evaluation uses two levels. The mechanism-level experiment checks whether the modules change internal variables such as disagreement score and posi-

tion size. The return-level experiment checks whether these internal changes affect simulated strategy return and drawdown. Both levels are necessary. A mechanism can look reasonable but have no effect on final decisions; conversely, a return difference can be accidental if the mechanism-level change is not observable.

All experiments are produced by local scripts in the project directory. The plots are generated from CSV and JSON outputs rather than manually drawn values. This improves reproducibility and reduces the risk of reporting inconsistent numbers.

5.2 Mechanism-Level Ablation

Table 2 shows the mechanism-level ablation result for the cached 000630 case.

Table 2 – Mechanism-level ablation result

Model	Total disagreement	Direction disagreement	Evidence disagreement	Suggested position
Ours-full	0.466	0.965	0.366	20.57%
w/o CW-RAG	0.473	0.965	0.403	20.42%
w/o DADM	0.000	0.000	0.000	30.00%
w/o RMSE	0.465	0.965	0.366	21.63%

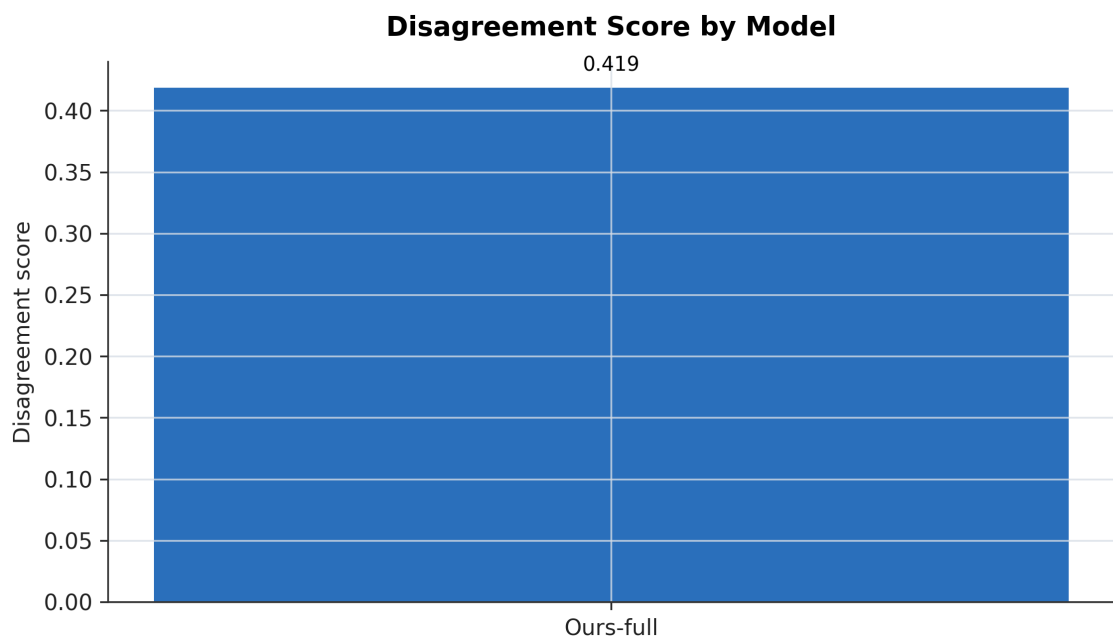


Figure 1 – Disagreement score comparison

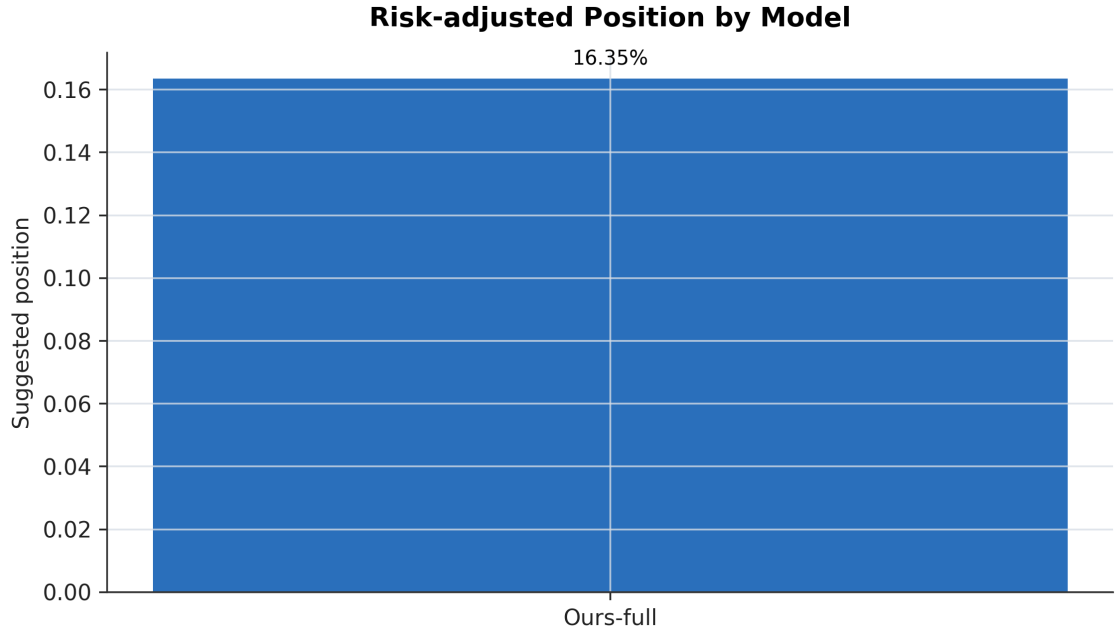


Figure 2 – Suggested position comparison

The full model detects a high direction disagreement and therefore limits the position. The model without disagreement diagnosis cannot observe the conflict and raises the position to 30.00%. This confirms that disagreement diagnosis is not only an explanation variable but also a risk-control variable.

The ablation result also shows the different roles of the three modules. Removing credibility weighting changes the evidence disagreement from 0.366 to 0.403. This means that evidence weighting affects how conflicting sources are interpreted. Removing review memory keeps the total disagreement close to the full model but increases the final position slightly. This indicates that review memory mainly works through calibration rather than through the current disagreement structure.

The most important contrast is between Ours-full and w/o DADM. The w/o DADM model reports zero disagreement, not because agents truly agree, but because the disagreement module is disabled. This is dangerous in a decision system: the absence of a measurement can be mistaken for the absence of risk. The full model avoids this problem by keeping the conflict visible.

5.3 Single-Case Return Simulation

For the 000630 case, the five-day realized stock return after the analysis date is -3.22%. If the model outputs buy, strategy return is computed as

$$R_{\text{strategy}} = pR_{\text{stock}}. \quad (15)$$

The results are shown in Table 3.

Table 3 – Single-case simulated return

Model	Action	Position	Stock return	Strategy return
Ours-full	Buy	20.57%	-3.22%	-0.66%
w/o CW-RAG	Buy	20.42%	-3.22%	-0.66%
w/o DADM	Buy	30.00%	-3.22%	-0.96%
w/o RMSE	Buy	21.63%	-3.22%	-0.70%

The full model loses less than the variant without disagreement diagnosis because it takes a smaller position in a high-conflict case.

The single-case simulation should not be overinterpreted. One stock and one date cannot prove trading profitability. Its value is diagnostic. It shows that, under the same negative future return, a higher position leads to a larger loss. Since the full model takes a lower position when disagreement is high, it suffers a smaller loss than the model without disagreement diagnosis.

This case also illustrates why position control matters. If two models produce the same buy action, they may still have different risk profiles. A binary action label hides this difference, while position size exposes it. Therefore, the thesis evaluates both final action and suggested position.

5.4 Four-Stock Backtest

The batch backtest uses four stocks and 20 samples. The performance metrics are total return, average return, volatility, maximum drawdown, win rate, and a Sharpe-like score:

$$S = \frac{\bar{R}}{\sigma_R + \varepsilon}. \quad (16)$$

Table 4 presents the backtest summary.

Table 4 – Four-stock backtest summary

Model	Samples	Total re- turn	Volatility	Max draw- down	Sharpe- like
Ours-full	20	-0.59%	0.72%	2.01%	-0.037
w/o CW-RAG	20	-4.37%	0.58%	4.37%	-0.383
w/o DADM	20	-7.33%	0.95%	7.68%	-0.394
w/o RMSE	20	-9.71%	1.51%	9.71%	-0.330

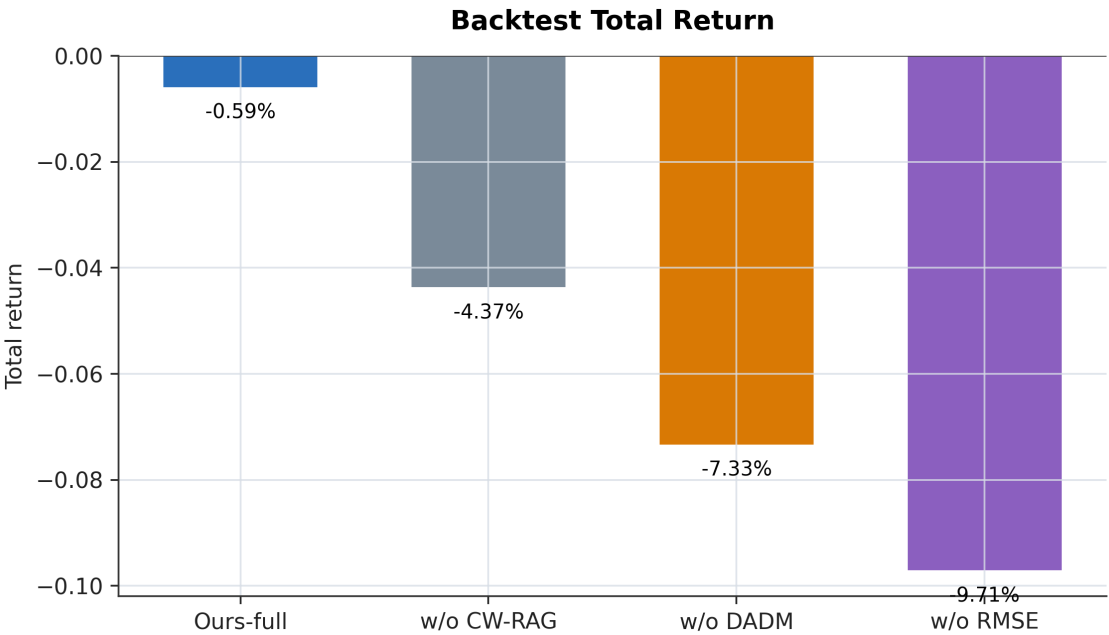


Figure 3 – Total return in the four-stock backtest

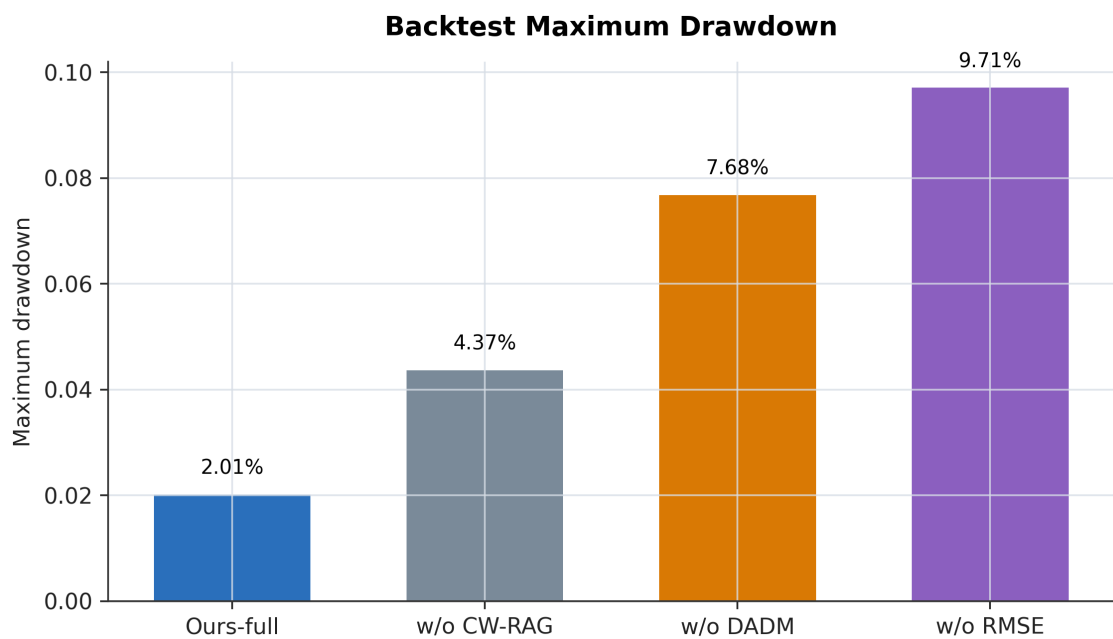


Figure 4 – Maximum drawdown in the four-stock backtest

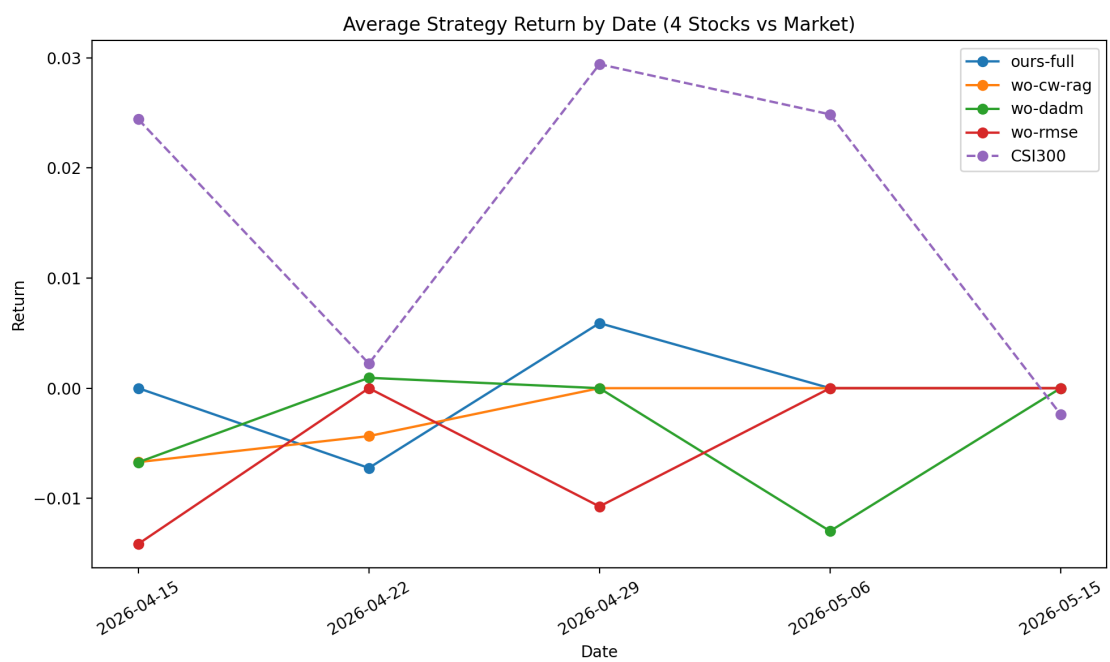


Figure 5 – Normalized strategy equity curve

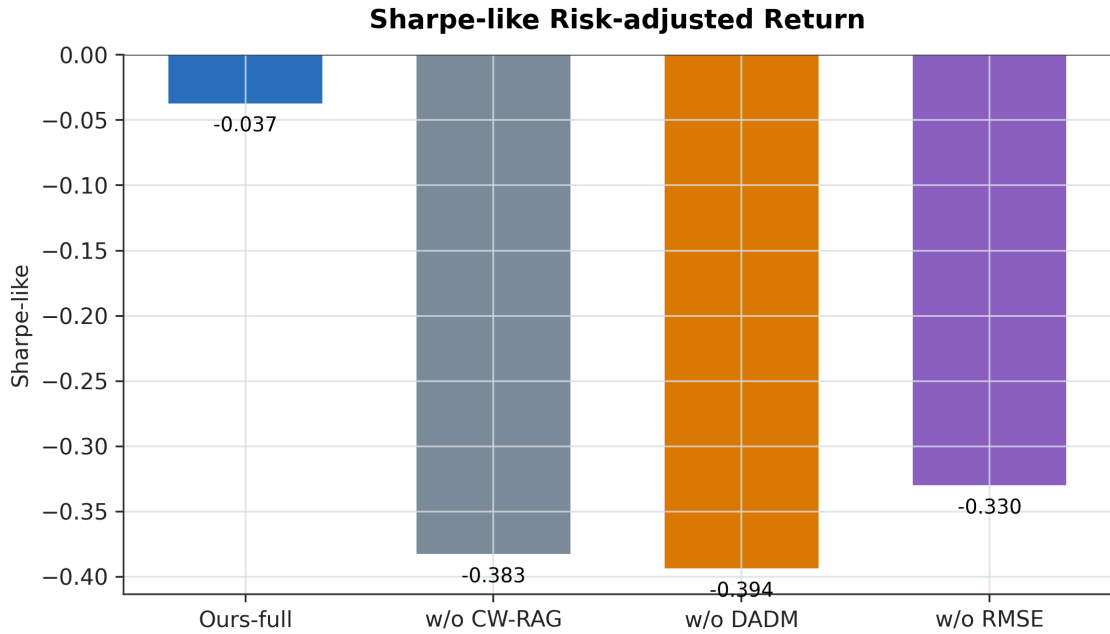


Figure 6 – Sharpe-like score in the four-stock backtest

The full model still has a small negative return in this short sample, but it has the lowest loss and the smallest maximum drawdown. This result supports the main hypothesis of the thesis: uncertainty-aware control may not guarantee profit, but it can reduce risk exposure when agent opinions conflict.

The maximum drawdown comparison is especially important. The full model has a maximum drawdown of 2.01%, while the ablated variants reach 4.37%, 7.68%, and 9.71%. This suggests that the complete uncertainty pipeline is more defensive in adverse samples. The Sharpe-like score is also less negative for the full model than for the other variants.

The equity curve uses the original project-generated line chart rather than a redesigned line chart. This preserves the raw visual style of the terminal output and makes the figure closer to the experiment artifact. The bar charts are formatted for readability, but the line chart is intentionally kept close to the original project output as requested.

5.5 Threats to Validity

There are several threats to validity. First, the backtest sample is small. A larger dataset covering different market regimes is required before making strong claims about investment performance. Second, the LLM-generated raw outputs may depend on prompts, model provider, and data availability. Third, the evidence credibility

weights are partly heuristic. Although the formulas are transparent, the initial weights may not be optimal.

Fourth, the strategy simulation is simplified. It does not include transaction costs, slippage, liquidity constraints, limit-up and limit-down rules, or portfolio-level capital allocation. Fifth, the system evaluates individual stock decisions rather than a complete portfolio. These limitations do not invalidate the mechanism-level finding, but they limit the financial strength of the backtest conclusion.

5.6 Reproducibility

The experiment can be reproduced from the project scripts. The cached ablation uses `run_ablation_from_cache.py`. The batch backtest uses `run_backtest.py`. The paper figures are generated by `plot_paper_figures.py`. The main result files are `ablation_results.json`, `backtest_summary_4stocks.csv`, and `backtest_results_4stocks.csv`.

The thesis uses these generated files directly. Therefore, the numerical tables and figures are consistent with the project outputs. If new model outputs or new price data are used, the figures can be regenerated by running the plotting script again.

6 Conclusion

This thesis designed and implemented a disagreement-aware multi-agent investment research decision system based on Trading-Agents-OpenClaw. The system introduces credibility-weighted evidence retrieval, multi-dimensional disagreement diagnosis, adaptive position control, and review-memory calibration. The formulas were defined explicitly so that evidence weights and disagreement weights satisfy proper normalization constraints.

The experiments show that disabling disagreement diagnosis causes the system to ignore agent conflict and increase position size. In the four-stock backtest, the complete system achieves the smallest loss and drawdown among the compared configurations. The result indicates that the proposed approach improves interpretability and provides useful risk control in uncertain financial environments.

The work still has limitations. The backtest sample is small, the evidence scoring weights are partly heuristic, and the review memory mechanism is still rule-based. Future work should extend the system to larger stock pools, portfolio-level allocation,

stronger risk measures such as VaR and CVaR, and more rigorous benchmarks for financial LLM agents.

Several future directions are promising. First, the evidence scoring model can be learned from historical review data rather than fixed manually. Second, the system can add market-regime detection so that memory from a bull market is not blindly reused in a bear market. Third, the framework can be extended from single-stock recommendations to portfolio construction. Fourth, the risk manager can incorporate more formal risk measures, including value at risk, conditional value at risk, and dynamic stop-loss rules. Finally, the evaluation should include factuality, citation quality, robustness, and explanation quality in addition to return metrics.

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